



V-Guard Sentinel

The inverter already knows when the power goes. Sentinel teaches it what to keep on and when its battery will quit.

Track 4 · From Smart Products to Intelligent Products

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Finale · Kochi · 24 September 2026



10:03 pm. The battery dies with no warning.

Three hours into a power cut, Home 6 in our village goes dark. Home 1 has the same battery and appliances and stays lit.



85 %

of Indian households face a power cut every day (*LocalCircles, 2023*)

×2 wear

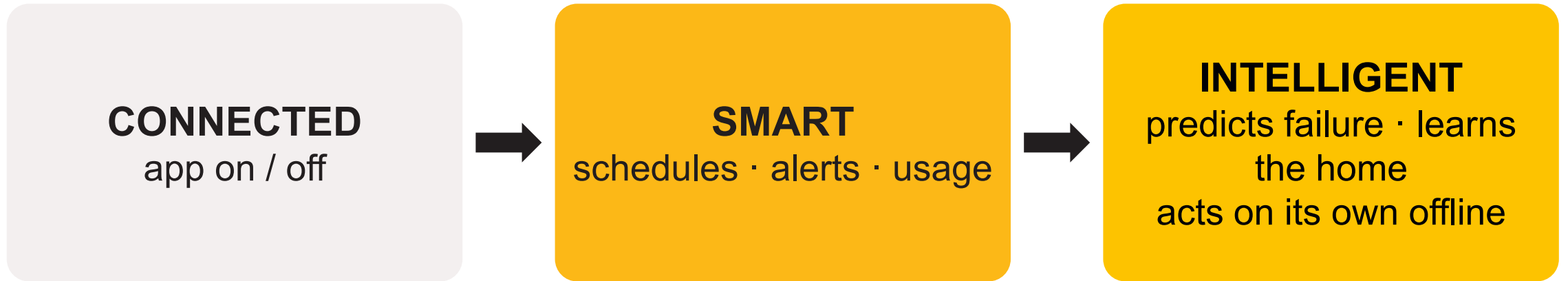
every +10 °C: lead-acid life halves with heat the Indian summer is the battery's enemy

≈ 1.5 %

of V-Guard revenue goes to warranty (≈ ₹69 cr, FY24)

Sentinel turns a smart inverter into an intelligent one.

Smart products take commands. Intelligent products predict, decide and act without the cloud.

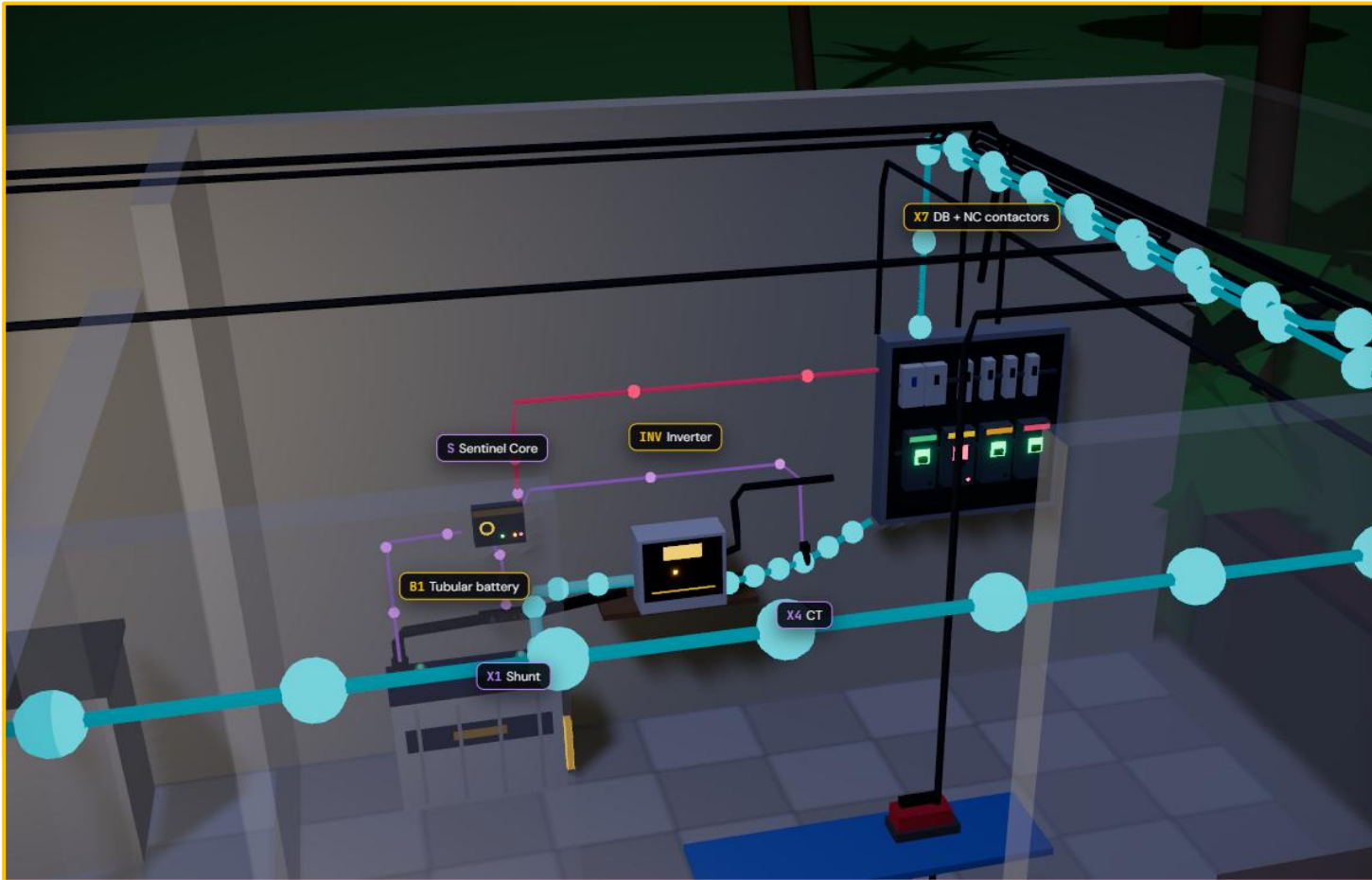


← Sentinel

- 1 Embedded** a daughter-board inside new V-Guard inverters
- 2 Retrofit** a 30-minute electrician fit on the installed base
- 3 Offline-first** every core function works with zero internet

One small box reads the battery and switches the loads.

Nothing on the battery or the inverter is modified on the retrofit only the negative cable passes through a shunt.



- 1 Shunt in the battery – lead**
battery current → INA228
- 2 NTC on the battery post**
temperature: the #1 ageing driver
- 3 CT + voltage tap on AC-OUT**
real & reactive power → ATM90E32AS
- 4 4 NC contactors in the DB**
coil off = load ON, always
- 5 ESP32-S3 Sentinel Core**
90 × 70 × 35 mm, powered by the battery

Built in, Sentinel costs about half as much.

Same core board. The embedded version reuses what the inverter already has.

RETROFIT

a box beside any inverter

Battery health
+ autopilot

₹1,050

+ Energy Coach
+ Grid Shield

₹1,650–2,050

Electrician fit, 30–45 min

EMBEDDED

inside new V-Guard inverters

Battery health
+ autopilot

₹450–700

+ Energy Coach
+ Grid Shield

₹850–1,300

Factory fit, and it can control the charger

Embedded reuses the inverter's own:

current shunt

12 V supply

enclosure

mains signal

Component cost at prototype volumes, from Indian distributor prices; contactors extra. Retail pricing is V-Guard's call.

Four jobs, one board & all on ESP32

1

Battery health

Predicts a replacement window: 'replace in 6–14 weeks'

2

Habit autopilot

Keeps the fridge, lights and router alive during a cut

3

Energy coach + grid shield

Names the big appliances; logs every sag and swell

4

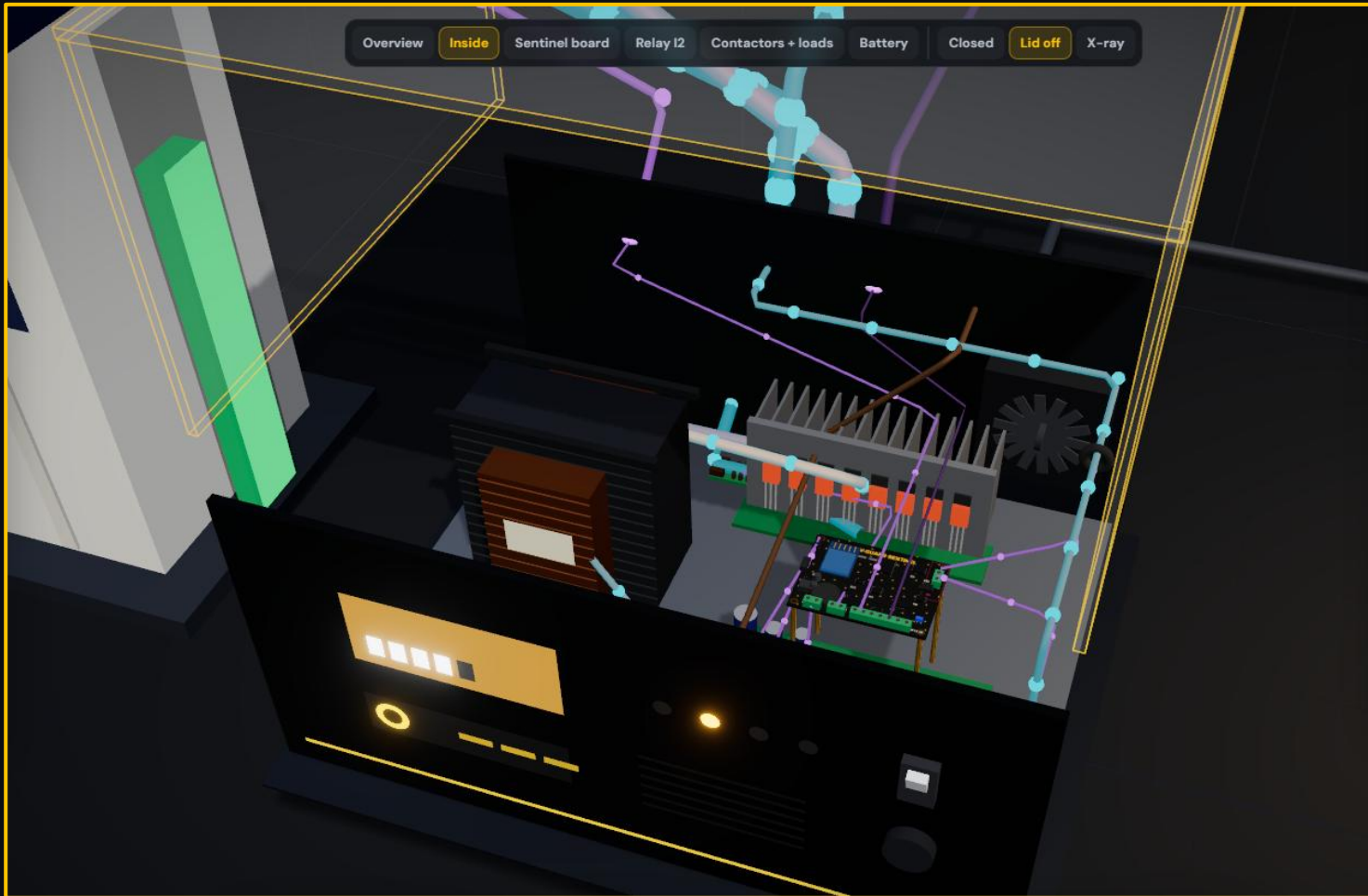
Signed health log

Tamper-proof record: warranty decided by evidence

Same core, same sensors, different models → the platform extends to pumps, stabilisers and water heaters.

Inside the inverter, Sentinel taps what's already there.

Our 3D proof of concept of the Embedded SKU: lid off, every part clickable, live power flow.

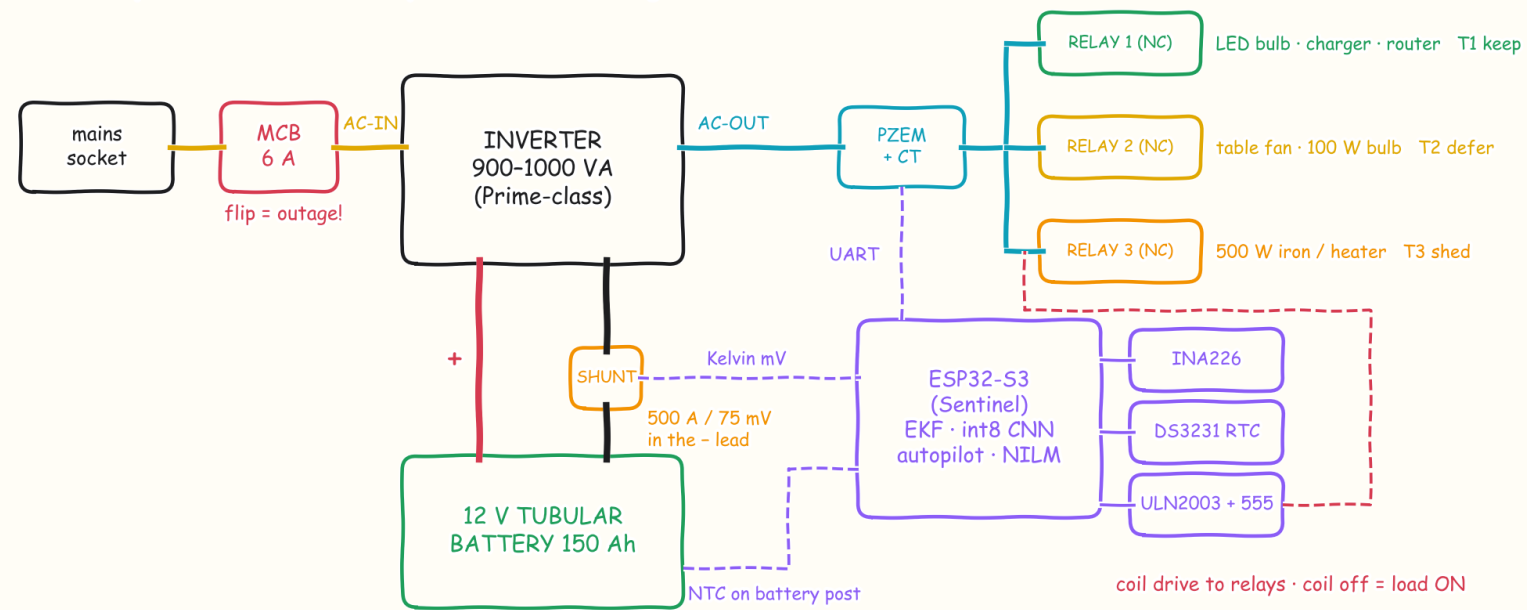


- 1 Inverter's own shunt (I8)**
→ INA228 · battery current
- 2 Mode / mains line**
→ PC817 opto
- 3 SG3525 charge feedback**
→ MCP4725 DAC · temperature-compensated charging
- 4 AC-OUT rail**
→ AMC1311 + ATM90E32AS · V, P, Q
- 5 4 coil outputs**
→ NC contactor panel · T1 / T2 / T3 / medical

And this is exactly how the bench is wired.

Real battery, real inverter, off-the-shelf parts, every part available from Indian distributors today.

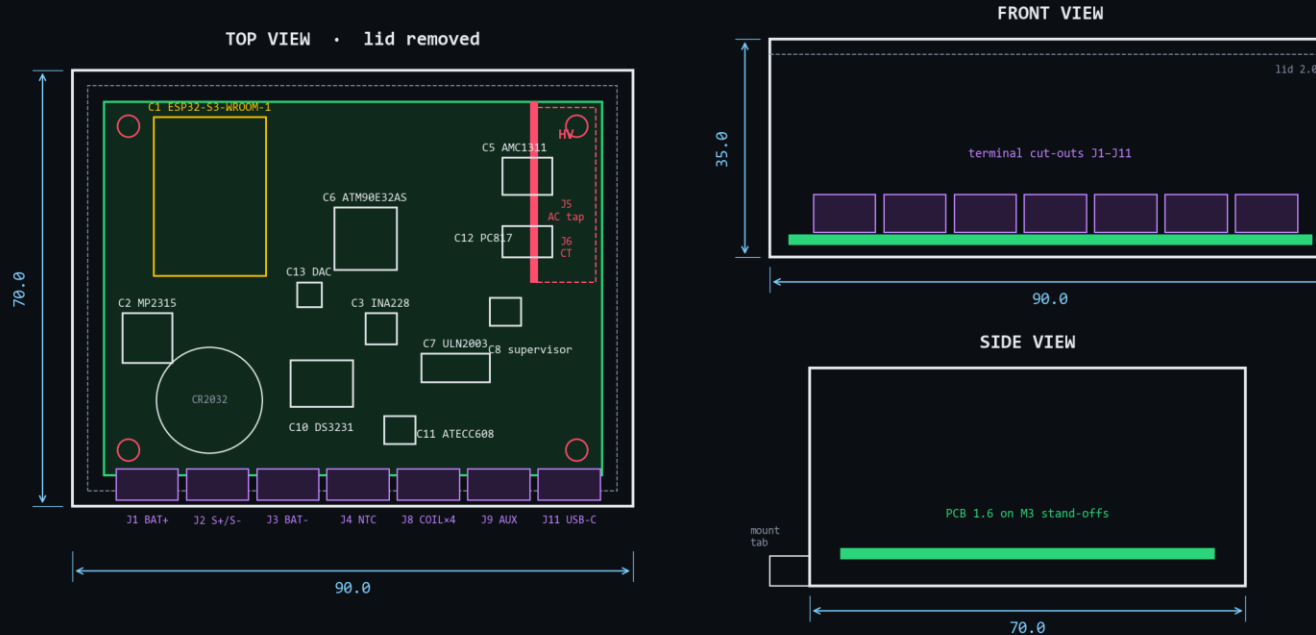
real battery · real inverter · three relay-switched tiers · nothing on the internet



Reset the ESP32 while T3 is shed → relay 3 re-closes in about a second.

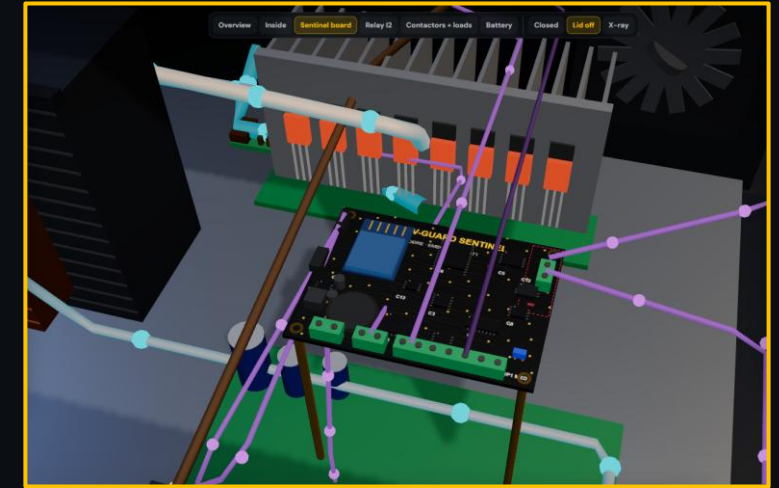
Designed to be built: 90 × 70 × 35 mm, HV zone fenced off.

Mechanical drawing from our round-3 spec, and the same board as a 3D render in the simulation.



V-GUARD SENTINEL CORE – ENCLOSURE & PCB

Enclosure 90 × 70 × 35 mm · PC/ABS UL94 V-0 · wall 2.5 · IP20
PCB 80 × 60 × 1.6 · 4× M3 (Ø3.5) at 4 mm inset
Battery current never enters the PCB – only shunt mV (J2)
AMC1311 + PC817 straddle the HV isolation moat
Coil off = load on: ULN2003 + supervisory timer + JP1
Units mm · per prototype/02-03 (round 3) · Rev T1



Battery current never enters the PCB
only the shunt's millivolts do.

Isolation moat

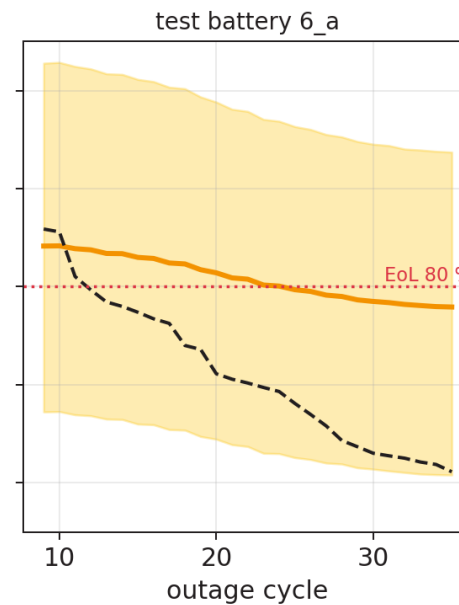
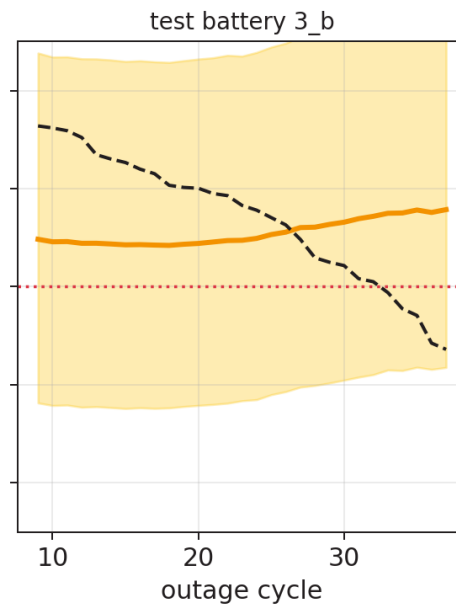
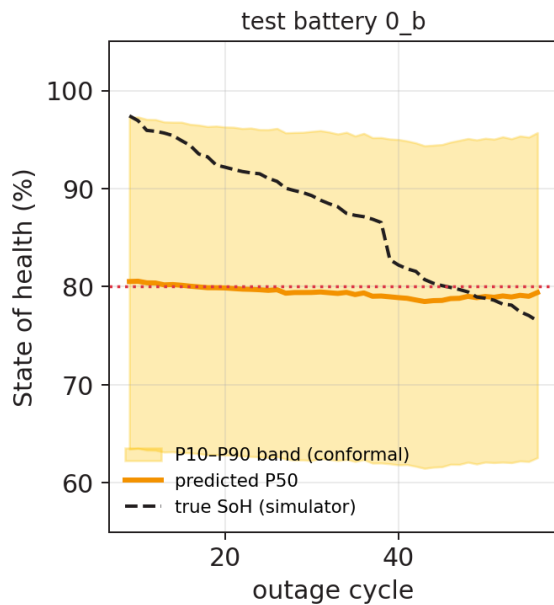
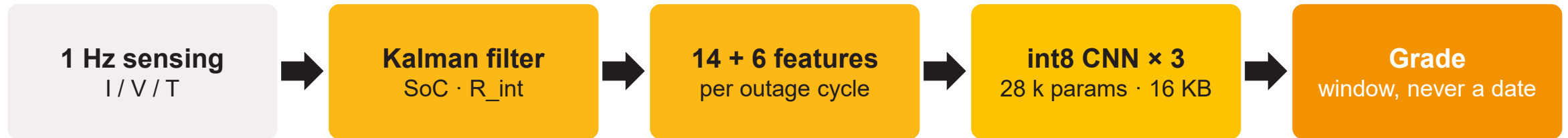
AMC1311 and PC817 straddle the HV zone.

UL94 V-0 PC/ABS, 4× M3

DIN or screw mount beside the inverter.

It gives a window, not a date: "replace in 6 –14 weeks".

Physics first, learning second every step runs on the ESP32.



8.2 pt

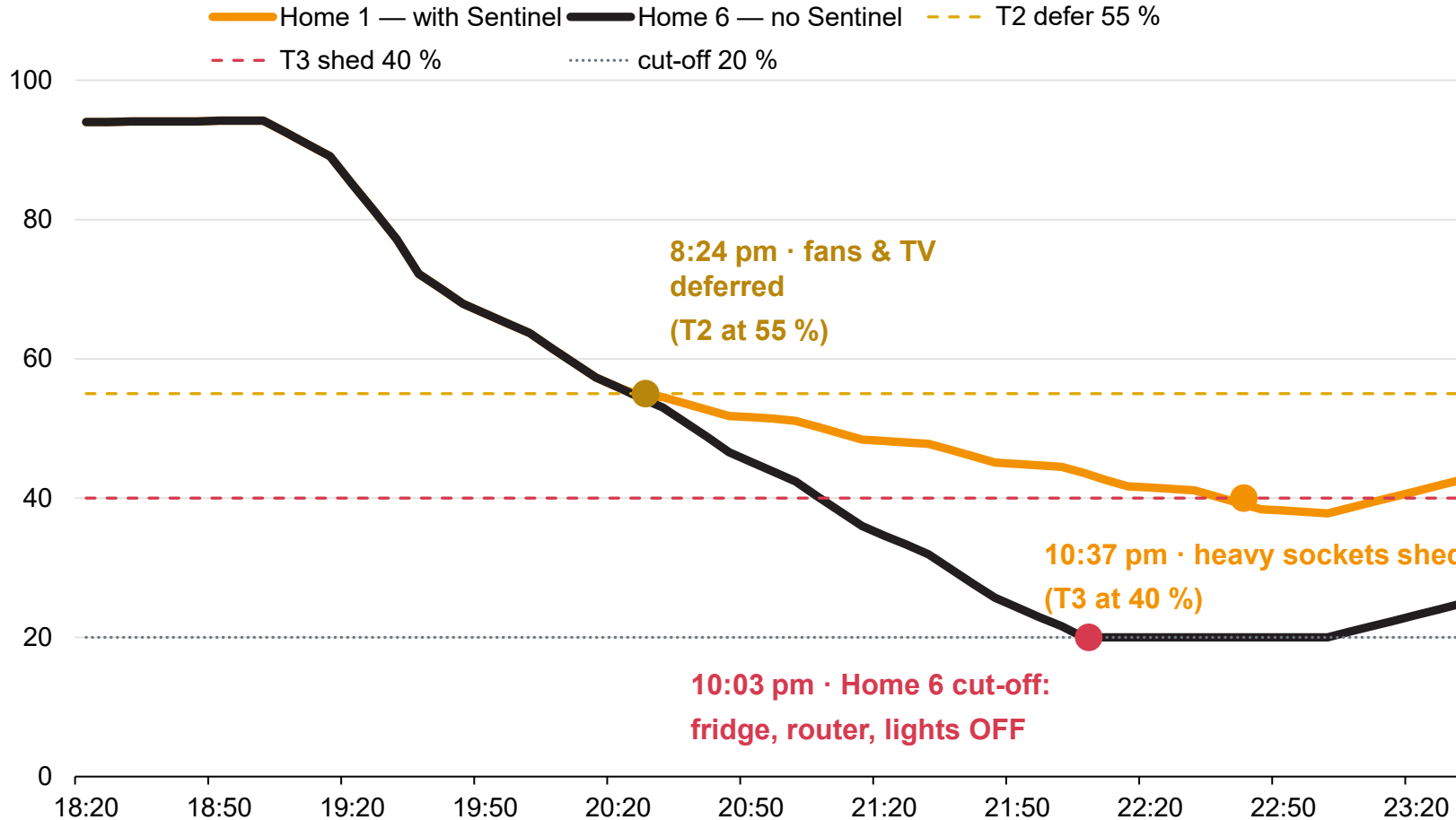
SoH error on 3 never-seen test batteries (Based on simulation data)

+0.01 pt

accuracy lost going float → int8 — it fits on the chip for free

Same outage, same battery: only Home 1 gets to keep its fridge on.

Battery charge through the 7–11 pm cut, straight from our simulation engine



57 min

Home 6 sat in the dark until mains returned at 11 pm

0 min

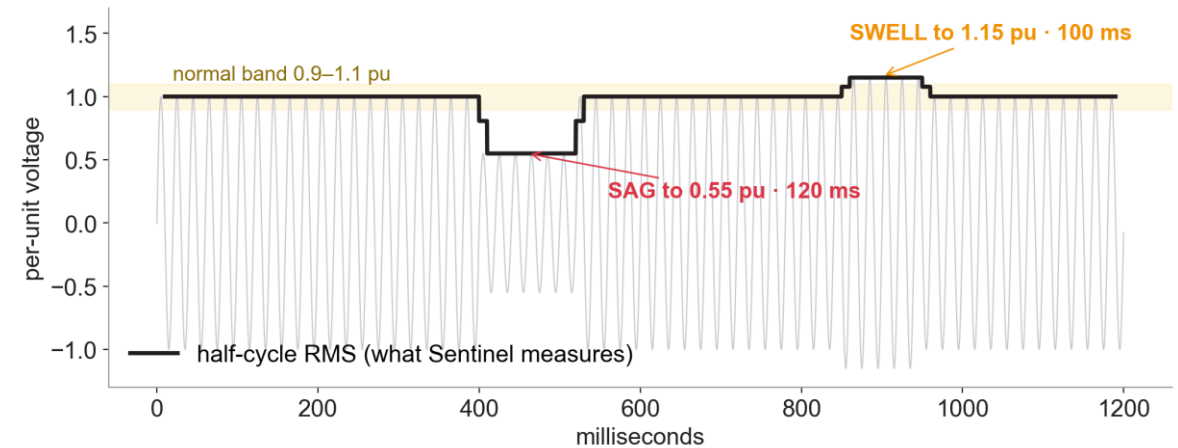
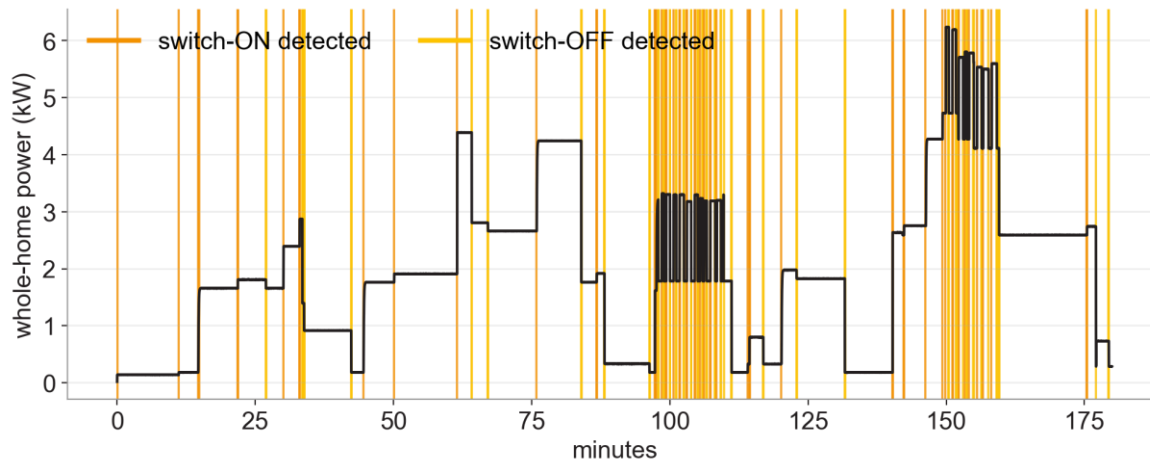
Home 1's fridge, router and hall light were never off

Coil off = load on.

NC contactors, a hardware supervisory timer and a medical jumper: any crash or reset brings every load back.

One sensor saves energy and catches a misbehaving grid.

Both run on the metering front-end (ATM90E32AS) and the isolated voltage tap (AMC1311)



Energy Coach 123 switch-ON/OFF events found in 3 h of whole-home power; each becomes a 13-number signature. Fridge F1 0.97, iron 1.00.

Grid Shield a 120 ms sag to 0.55 pu measured as 0.54; a 100 ms swell to 1.15 pu as 1.16. 13 / 13 dips caught. Logged, dated, signed.

Let's watch it happen live.

Three moments to watch for:



1 · The village

Feeder cuts at 7 pm. Watch Home 6 go dark at 10:03 while its twin stays lit.



2 · Flip the MCB

Relay swings in slow motion: 8 ms gap on the scope, outage confirmed in 1.5 s.

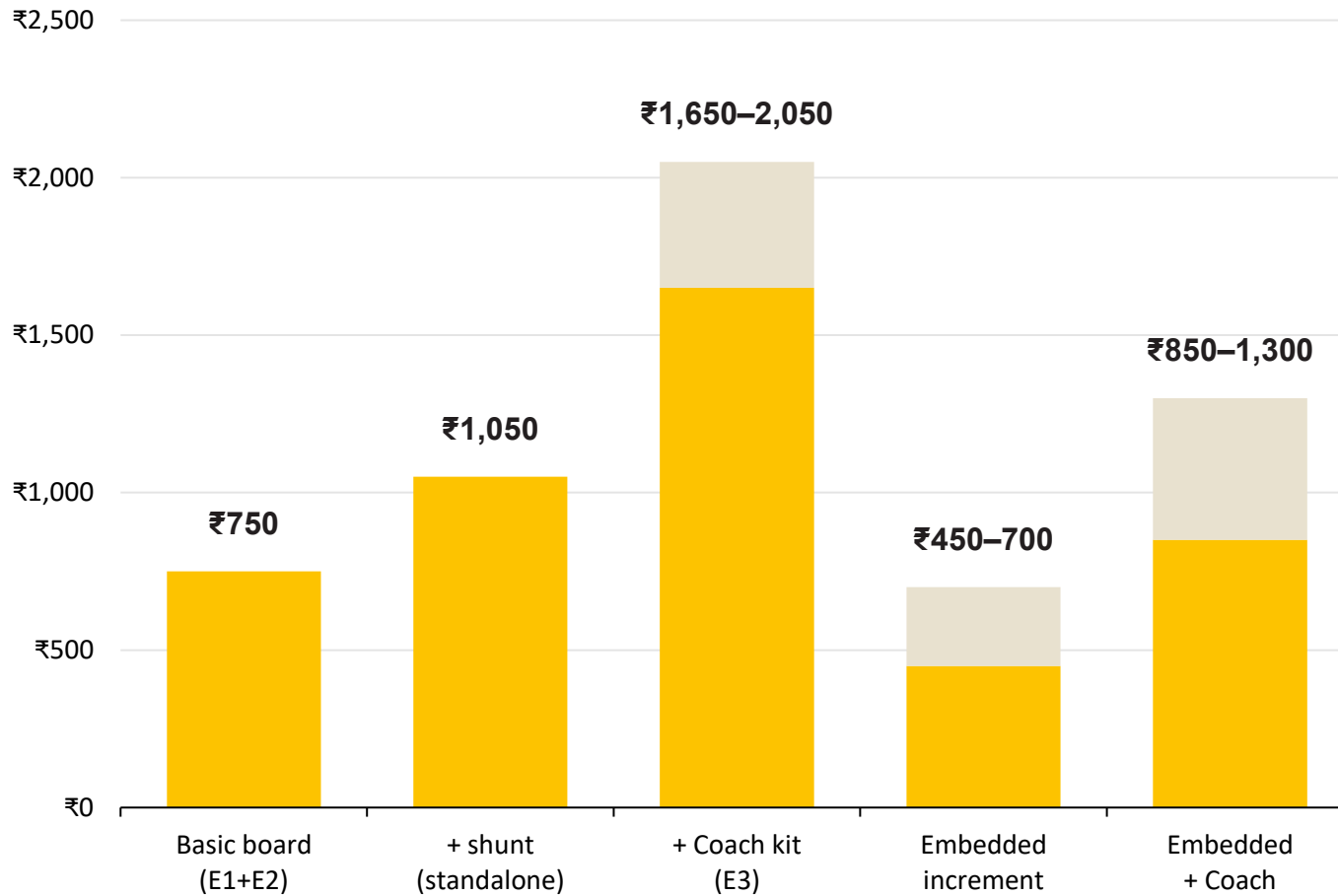


3 · Crash it

Drop SoC below 40 % → the iron sheds. Reset the chip → every load comes back.

Two SKUs, one board for 5–14 % of what one battery costs.

Cost stated by the engines each tier enables, against a ₹15 k tubular battery.



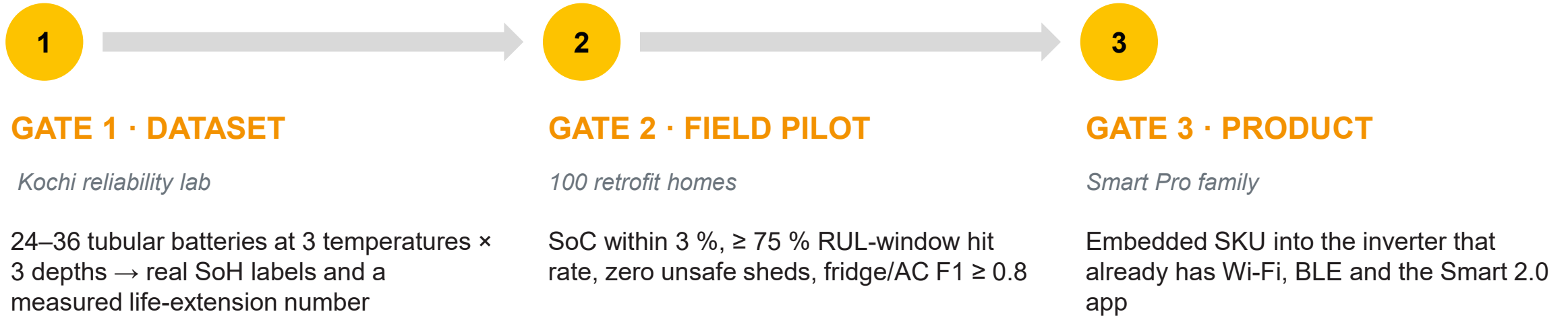
Sentinel-Embedded

Inside new V-Guard inverters. Taps the inverter's own shunt; controls the charger through a DAC with a hardware ceiling. Life-extension claims live here only.

Sentinel-Retrofit

Box beside any inverter. External shunt, NTC, CT and 4 fail-safe contactors. Advisory on charging, protective on loads. ≈ 2.6 mA average draw.

Three validation gates and what we need to pass the first.



What V-Guard gains

battery replacements at the right time · warranty by evidence · a premium 'intelligent' tier · a fleet-data moat

What we ask for

one opened Prime-series inverter + charger schematic · ESP32 bench parts · batteries and cyclers for Gate 1

**Coil off means loads on.
A window, never a date.
Offline first, fleet second.**

Thank you — we'd love your questions.
Team Codey Tingle (TI3271)





Isolation

reinforced AMC1311 barrier; HV zone behind a moat; TVS on every line

Fail-safe

NC contactors — any fault, reset or brown-out brings loads back

Standards

IS 13252 / IEC 62368-1 (BIS gate) · IEC 62040 UPS · IEC 61000 EMC · IEEE 1159 / IEC 61000-4-30 Class-S-like PQ · UL94 V-0 enclosure

Charging

-24 mV/°C compensation, derate above 45 °C, stop at 58 °C, 2 s heartbeat revert to factory